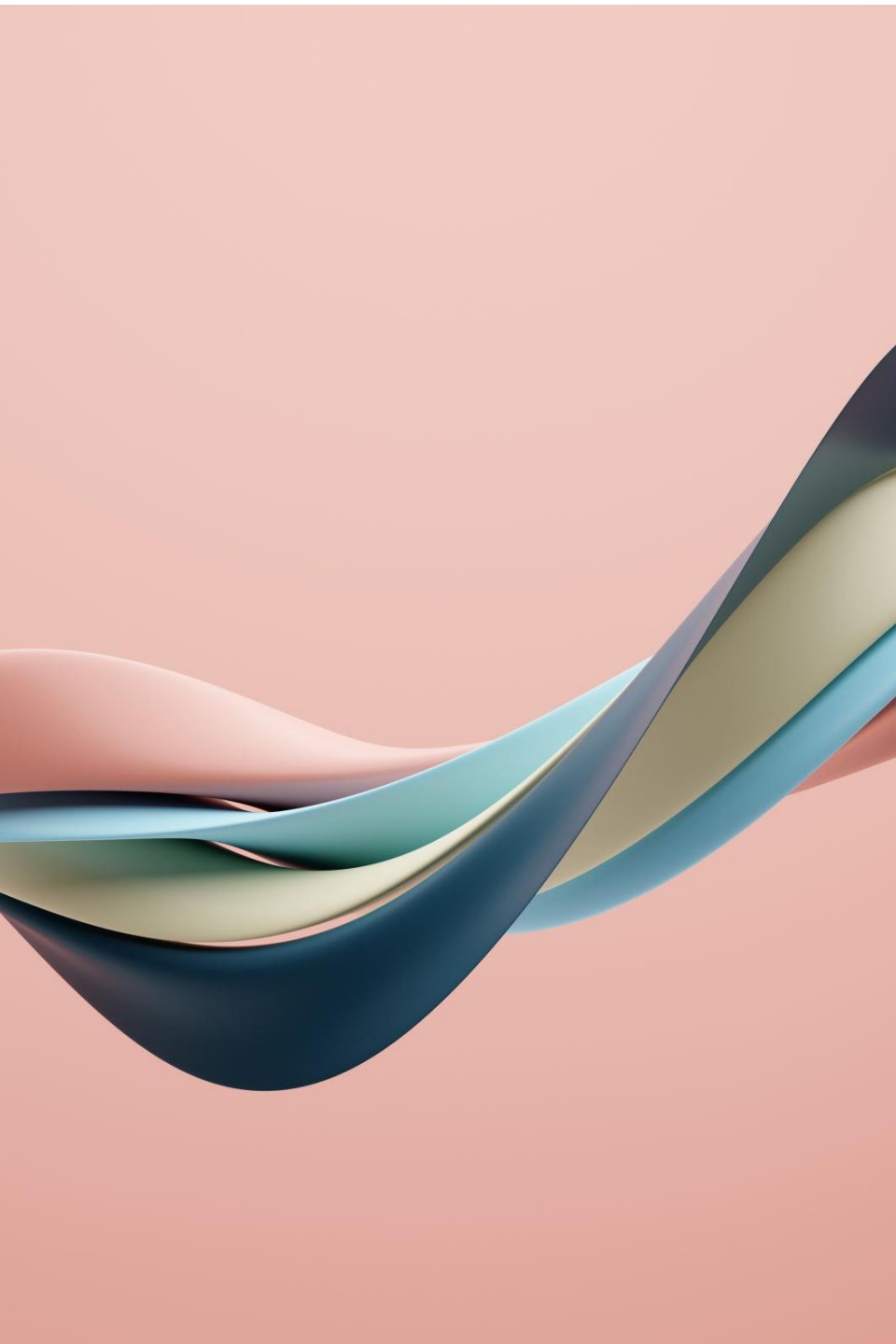




HCI-DE(M): THEORIES AND HEURISTIC EVALUATION

XIANGHUA (SHARON) DING
(XIANGHUA.DING@GLASGOW.AC.UK)

SOME THEORETICAL BACKGROUND



The DESIGN of EVERYDAY THINGS

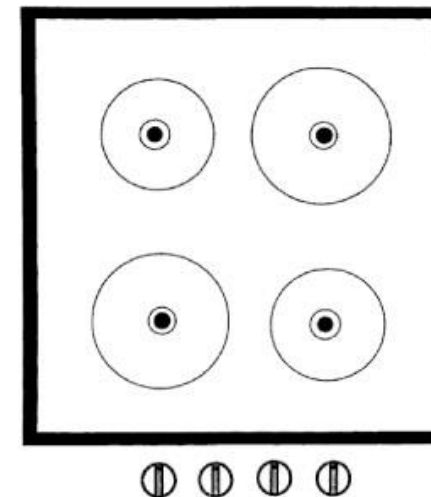
DON
NORMAN



**SOME ARE
EASY TO
USE**

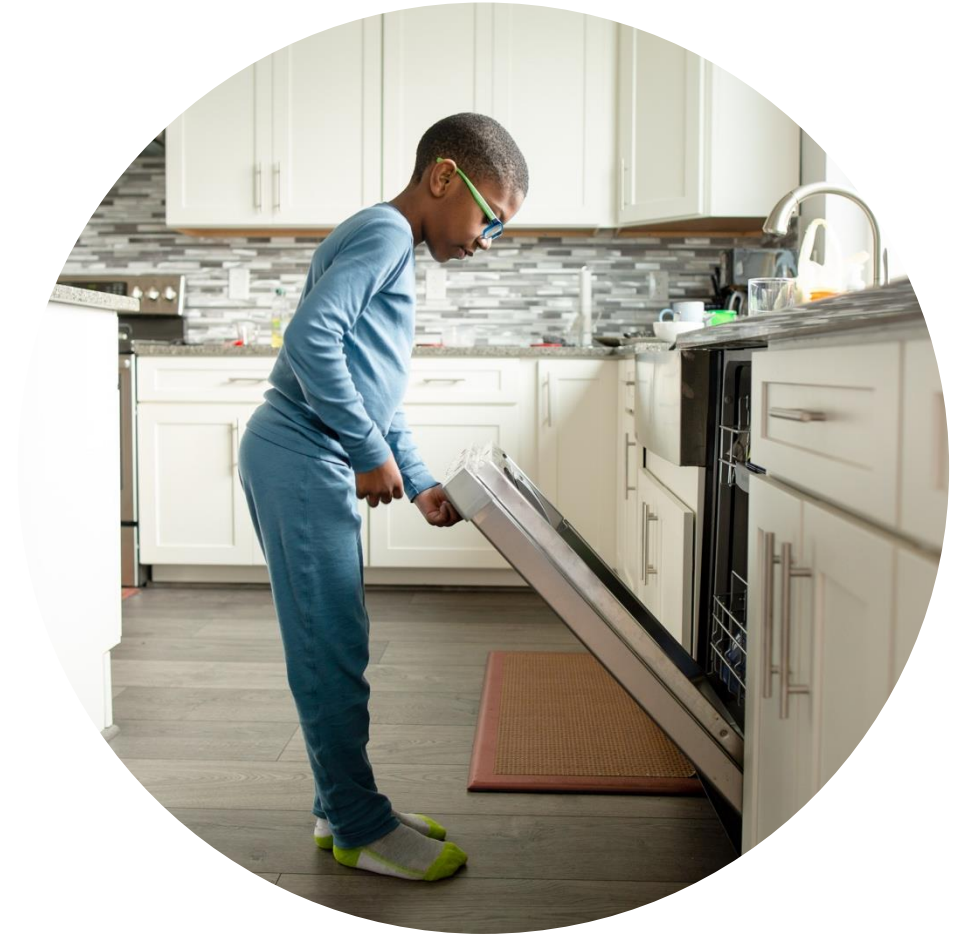


**SOME ARE
FRUSTRATING
TO USE**



EASE TO UNDERSTAND

WHEN WE FIRST SEE AN
OBJECT WE HAVE NEVER SEEN
BEFORE, HOW DO WE KNOW
HOW TO USE IT?



The background features a large, semi-transparent semi-circle on the right side, with a vertical bar on the left. The colors are muted greens, browns, and oranges.

AFFORDANCE

OBJECTS CONVEY STRONG CLUES TO THE OPERATION



WHICH IS EASIER TO KNOW HOW TO OPERATE?

AFFORDANCE



- The *relationship* between an object and a person, *not a property of an object*
 - Presence of affordance jointly determined by object's properties (materials, textures, size, height and other physical properties) and person with capabilities and past experiences
 - A chair affords sitting



- Prevention of actions

ANTI-AFFORDANCE

EXERCISE

- What properties invite the action of hanging laundry?
- What if we want to discourage that action?



WHAT IF AFFORDANCE CANNOT BE EASILY PERCEIVED

- Need to signal with a *signifier*
 - Image, text, sound...
 - Deliberate, e.g. labels
 - Emergent
 - Paths worn onto ground
 - Queues of people



EXERCISE

- The total misperception of affordance here. How shall we signal?



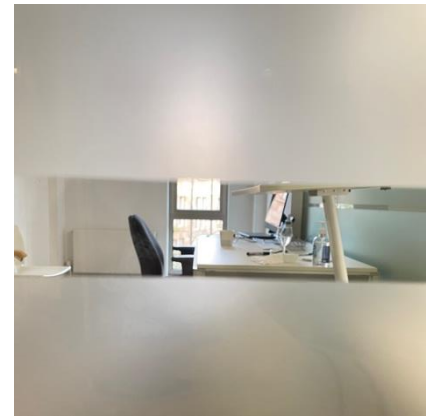


EXERCISE

- Look at this picture, and think about what actions it wants to support and what actions it wants to prevent? How is that information conveyed?

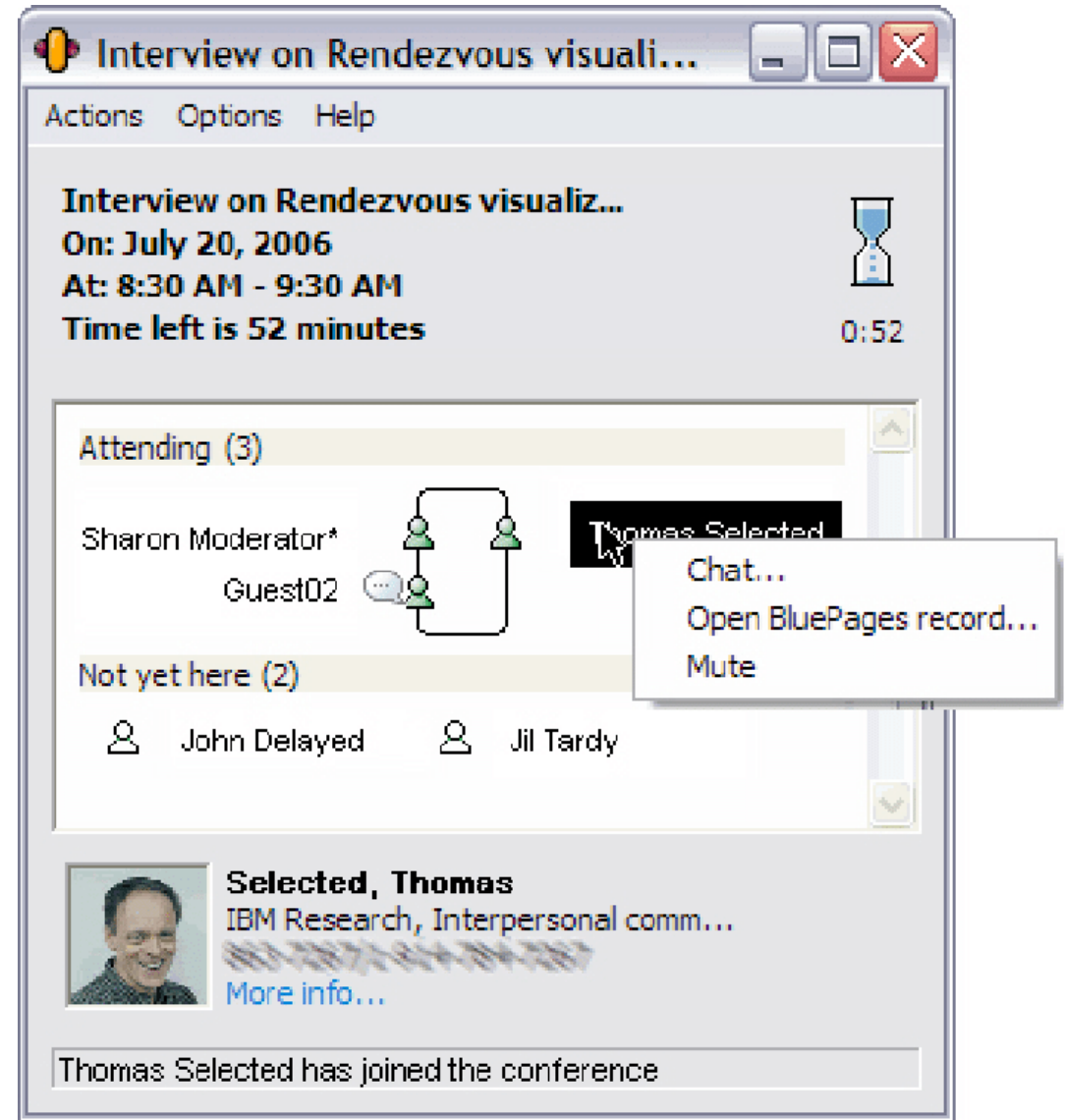
EXERCISE

The office wall: what actions we would like to support and what actions we would like to discourage?



EXERCISE

How about the virtual conference interface?



Social translucence in HCI

EASE TO USE

HOW CAN WE DESIGN
THINGS TO REDUCE THE
COGNITIVE LOAD, SUCH AS
MEMORIZATION, NEEDED
FOR CORRECT USE



KNOWLEDGE IN THE HEAD AND KNOWLEDGE IN THE WORLD

EVERYDAY ACTION

- Precision, accuracy and completeness of knowledge is not required for precise behavior
 - As long as we can recognize the difference



KNOWING WHAT TO DO

- Knowledge in the world
 - Perceived affordances, controls & their actions
 - Natural and cultural norms conditioning our action
- Knowledge in our heads
 - Memories and experience



KNOWLEDGE IN THE HEAD

- Short term memory (STM)
 - invaluable for everyday tasks
 - conscious processing
 - fragile: easily get distracted
 - capacity: for only 7+/-2 items
 - duration: for only about 30 seconds

714-865-2189

665-013-1124

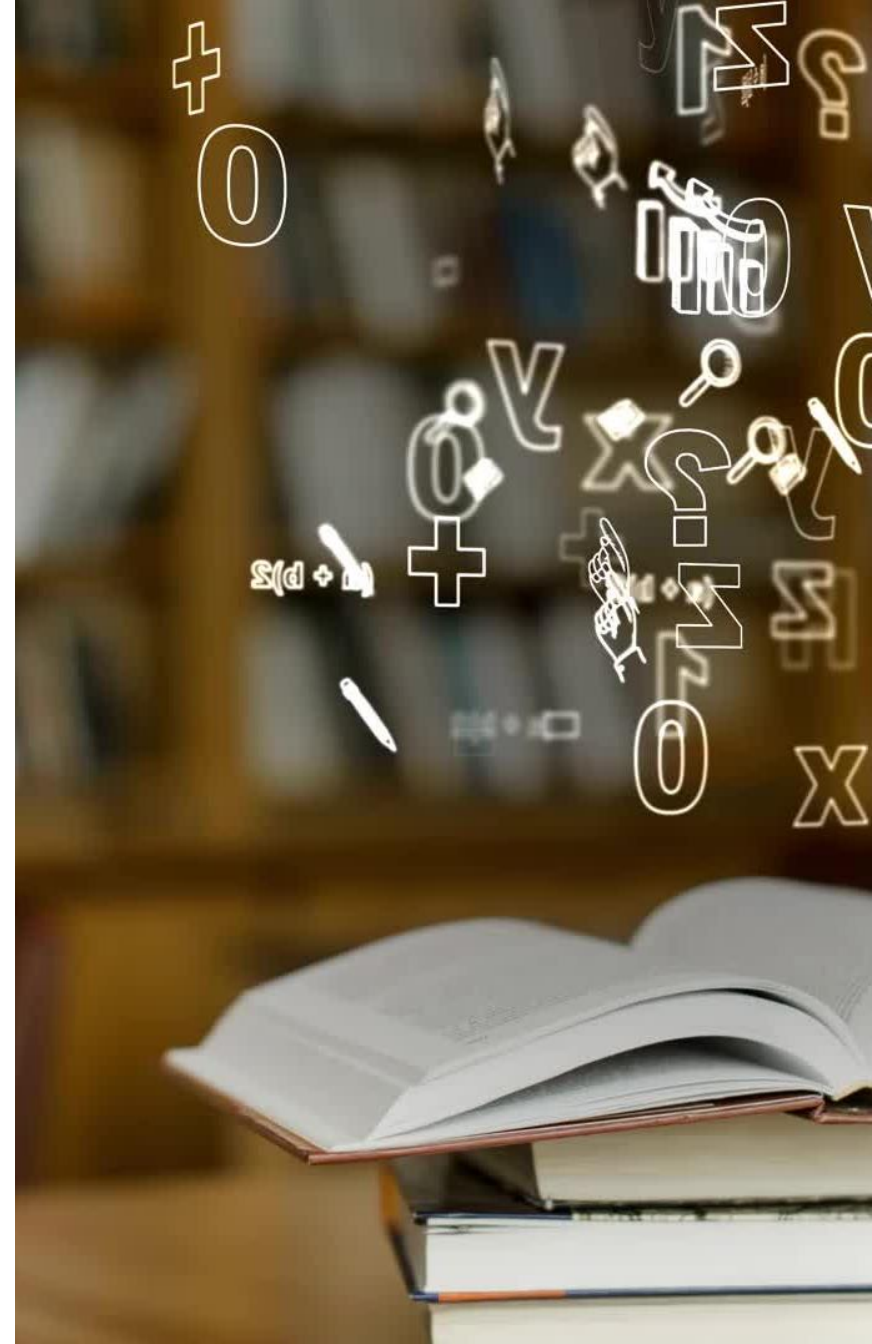
315-964-0281

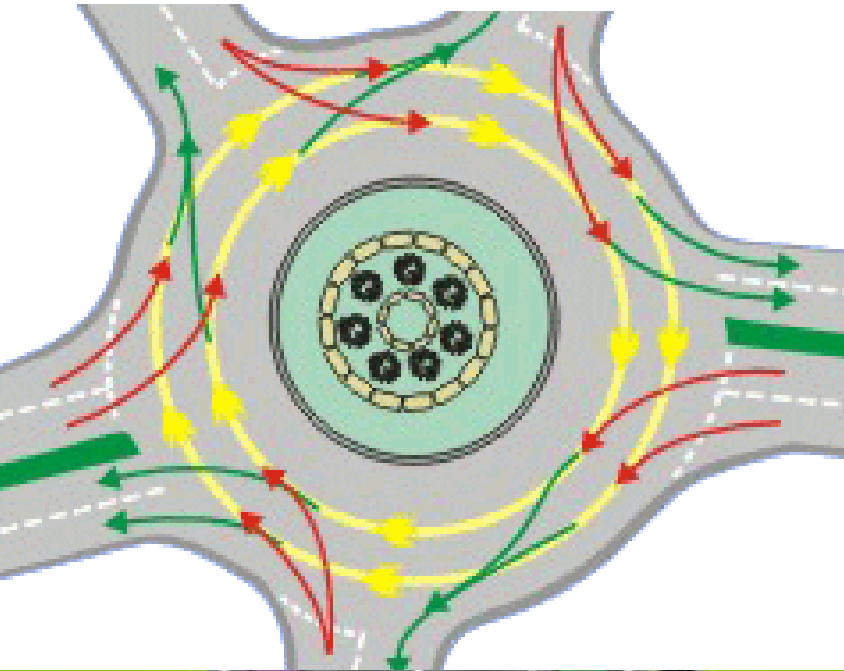
Making things into 7 items is a way to make it more manageable

KNOWLEDGE IN THE HEAD

- Long term memory (LTM), ways to maintain and retrieve our experience
 - memory of arbitrary things
 - memory of meaningful things

Storage and retrieval is easier when the materials make sense

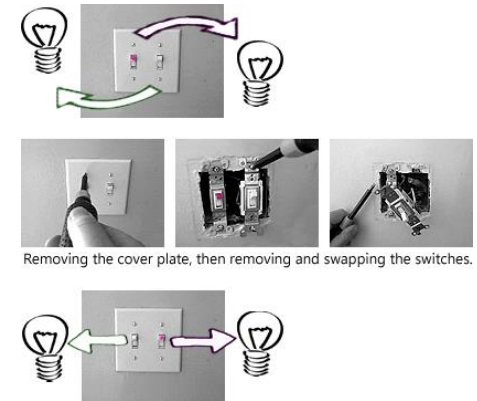




THE TRADEOFF

- knowledge in the world or knowledge in our heads (e.g. long term memory)?

THE PRINCIPLE OF MAPPING



- Forming natural mapping between things

THE PRINCIPLE OF FEEDBACK



- Showing the effect of an action

ERROR FREE

**HOW CAN WE AVOID
MISTAKES?**



TO ERR IS HUMAN

- slips vs. mistakes
- Assume all possible mishaps will happen and protect against it.



EXERCISE

- What error is this? Slip or mistake?
How shall we avoid it?

Role	Score	Review	Action	Invitation
1AC	0	review 4	remove edit	invite
2AC	0	review 1	remove edit	invite set role to 'Reviewer'
Reviewer	0	review 2	remove edit	accepted
Reviewer	0	review 3	remove edit	accepted

Omg it finally happened and I hit the REMOVE REVIEWER button (WHY IS THIS BUTTON EVEN THERE) and removed XXXX (the 2AC) from submission XXXX. It looks like it just removes the person and doesn't even ask you if you are sure.

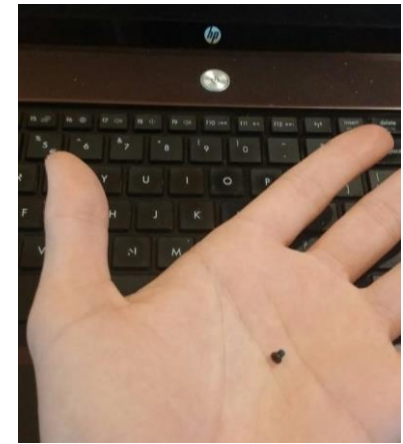
THIS IS A TERRIBLE INTERFACE PLEASE FIX ALL THINGS ABOUT IT.

1. Put XXX back on the paper!
2. Oh did it delete the review??? THAT WOULD BE HORRIBLE I don't even want to check.
3. Get rid of this terrible "remove" link.
4. Make it so that it asks you if you really want to remove someone.

We are CSCW, we really should have a better interface for all this.

CONSTRAINTS

THE POWER OF CONSTRAINTS



- Design to constrain choices: many possibilities could be associated with use difficulty
 - Physical Constraints – rely on properties of the physical world
 - Cultural Constraints – rely on socially learned behaviours
 - Semantic Constraints – rely on intrinsic meaning
 - Logical Constraints – rely on trial and reasoning

CONSTRAINTS TO FORCE FUNCTIONS

- Forcing Functions
 - Interlocks – Requiring actions to occur in sequence
 - Lock-Ins – Keeps an action active, preventing action from stopping
 - Lock-Outs – Prevents an action from occurring (typically in safety context)



EXERCISE

The case of Cash Machine:

What are the forcing functions in cash machines? Interlocks? Lock-ins? Lock-outs?



FORCING FUNCTIONS AND USABILITY

* New password ⓘ

✖ Your new password cannot be same as old password.

* Re-enter new password

Continue

- Balance error prevention with ease of use

DESIGN FOR ERROR

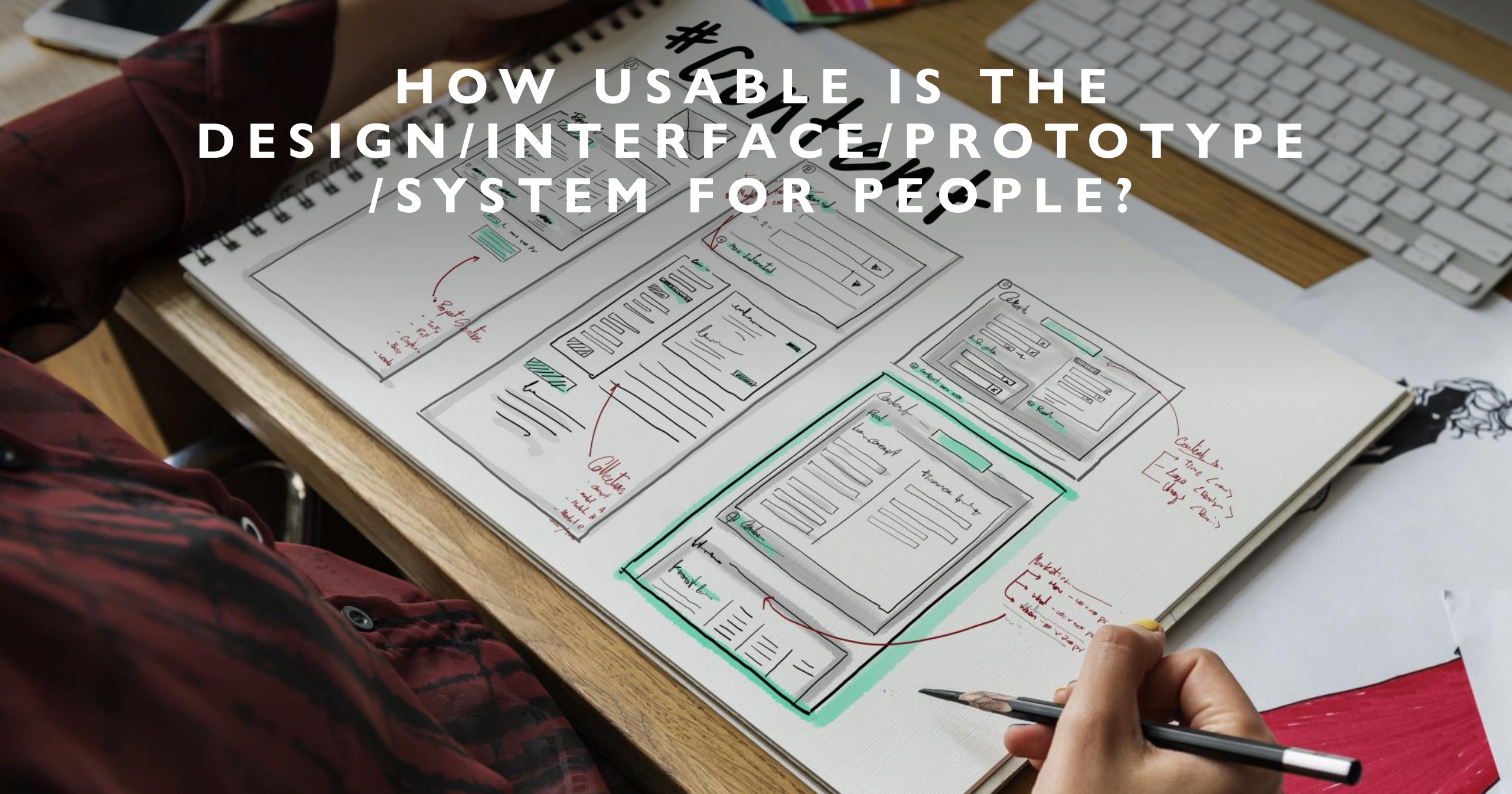
- change the attitude towards errors
- understanding the causes of error and design to minimize those causes
- make it easier to discover the errors that do occur, and make them easier to correct
- make it possible to reverse actions
- forcing functions



HEURISTIC EVALUATION

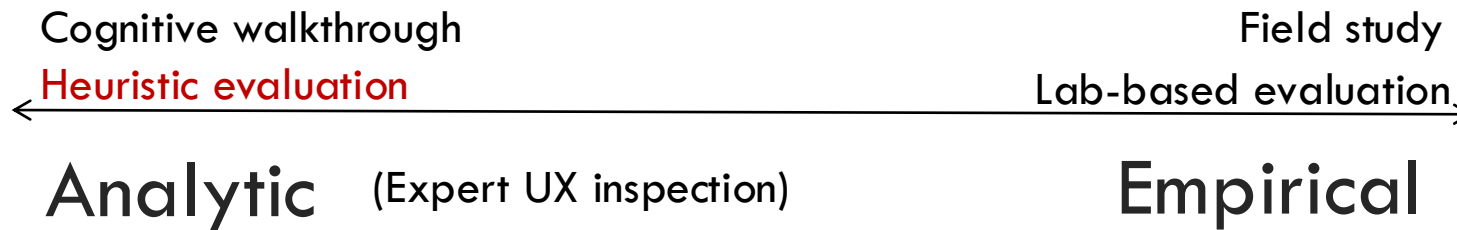


HOW USABLE IS THE DESIGN/INTERFACE/PROTOTYPE /SYSTEM FOR PEOPLE?



- Analytic
 - Based on looking at inherent attributes of the design, not seeing design in use
- Empirical
 - Based on data observed in the performance of real user participants

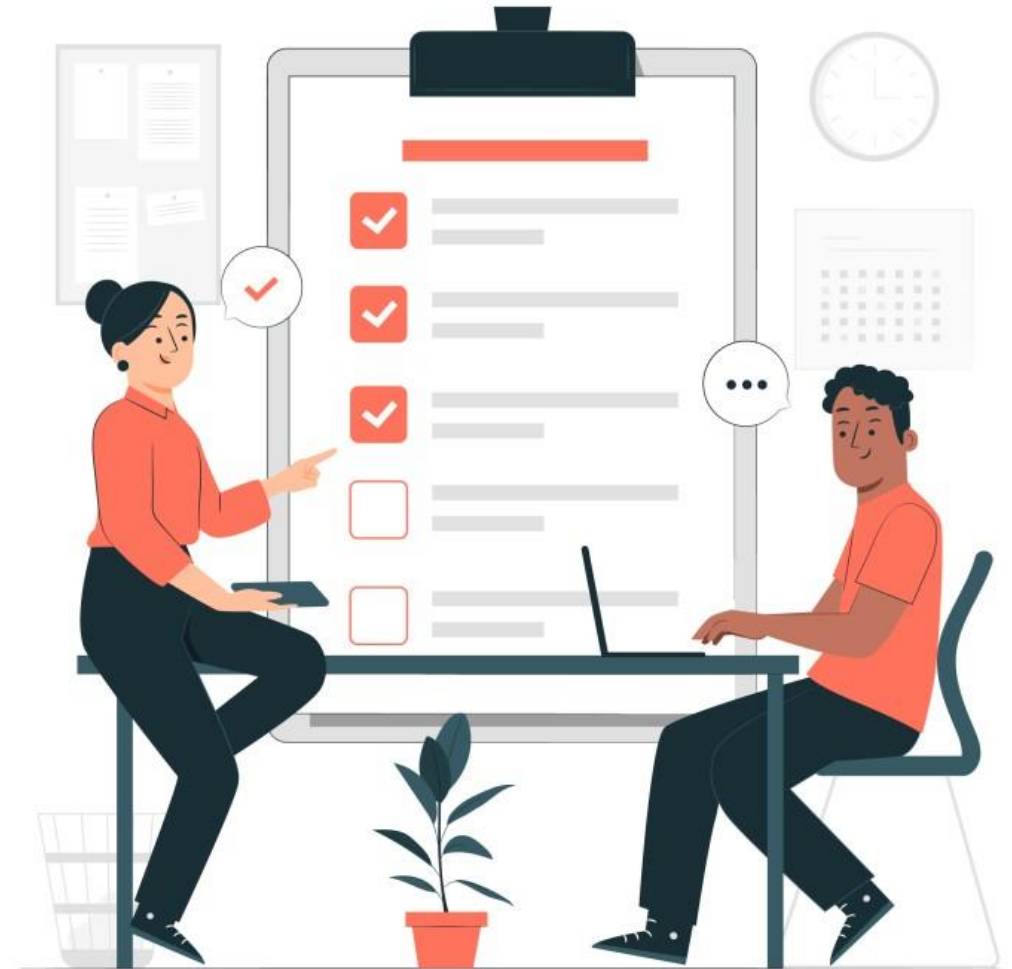
“If you want to evaluate a tool, say an axe, you might study the design of the bit, the weight distribution, the steel alloy used, the grade of hickory in the handle, etc., or you might just study the kind and speed of the cuts it makes in the hands of a good axeman”



ANALYTIC VS. EMPIRICAL METHODS

HEURISTIC EVALUATION

- Heuristic evaluation involves having a small set of evaluators examine the interface and judge its compliance with recognized usability principles (the "heuristics")

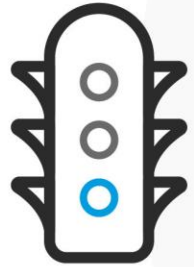


WHY IS THIS IMPORTANT?

- Foundational principles of usability
- Apply in many contexts
- Concrete and precise language
- Give you the tools to evaluate interfaces and improve your own interfaces



10 Usability Heuristics



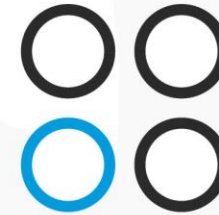
**Visibility of
System Status**



**Match Between System
& the Real World**



**User Control
& Freedom**



Consistency & Standards



Error Prevention



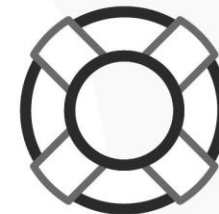
**Recognition Rather
than Recall**



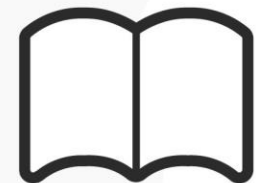
**Flexibility &
Efficiency of Use**



**Aesthetic &
Minimalist Design**



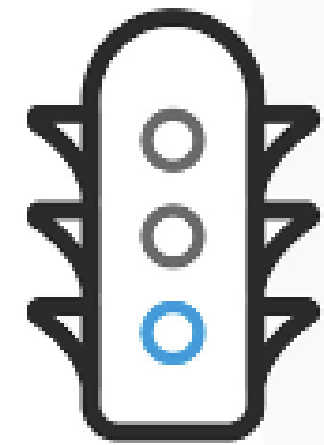
**Help Users Recognize, Diagnose
& Recover from Errors**



**Help &
Documentation**

1. VISIBILITY OF SYSTEM STATUS

- The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.



**Visibility of
System Status**

VISIBILITY

- Make it more visible in terms of what actions to support and how
 - What information dose the user need to know: balance it with information overload
 - How will users make decisions based on the information

Grade

☐ Quick grading ☒ Download submissions in folders Actions ▾

Reset table preferences

Select	First name / Last name	Username	ID number	Status ↑	Group	Grade	Last modified (submission)
☒							

Extend
Change marking state
Message
Share file with all selected users

Lock Unlock Download Delete More

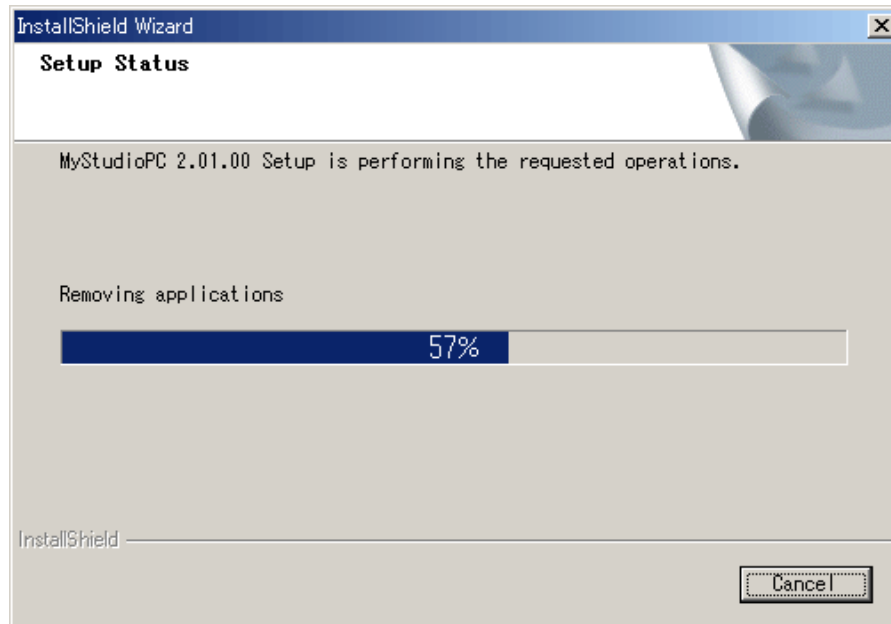
247 selected

Have you ever had experience that you clicked something but nothing happened? How did you feel? What did you do?

QUESTION

FEEDBACK

- To show the effect of an action:
 - Continuous internal states, progress through a task, or discrete actions
 - Will the user feel confident that their input was detected?



2. MATCH BETWEEN SYSTEM AND THE REAL WORLD

- The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.
- Follow real-world conventions, making information appear in a natural and logical order.



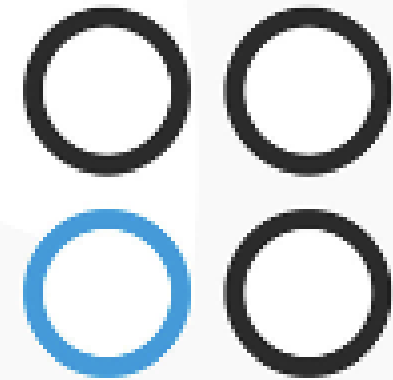
**Match Between System
& the Real World**

Donald Norman's book title was changed from "The Psychology of Everyday Things" to "The Design of Everyday Things". Why?

QUESTION

3. CONSISTENCY AND STANDARDS

- Users should not have to wonder whether different words, situations, or actions mean the same thing
- Recall internal and external consistency in terms of layout languages



Consistency & Standards

How did you learn to use google spreadsheet?

QUESTION

4. RECOGNITION RATHER THAN RECALL

- It's easier to recognise something from a prompt rather than recall in isolation
 - Minimize the user's memory load by making objects, actions, and options visible.
 - The user should not have to remember information from one part of the dialogue to another.
 - Instructions for use of the system should be visible or easily retrievable whenever appropriate



**Recognition Rather
than Recall**

How can you improve this?

Your Address *

Street Address

Address Line 2

City

State / Province / Region

Postal / Zip Code

Country

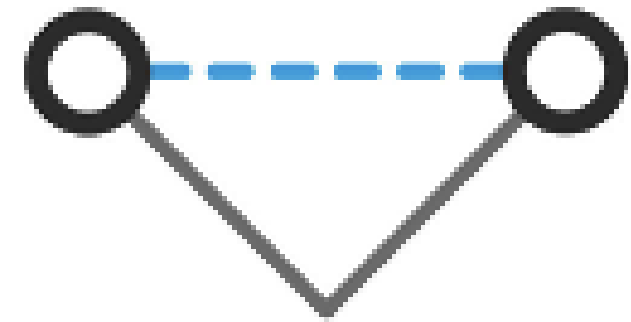
Email Address

Phone Number

QUESTION

5. FLEXIBILITY AND EFFICIENCY OF USE

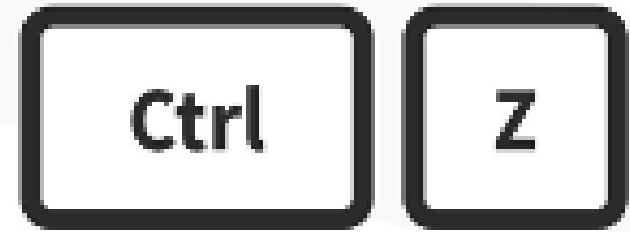
- Accelerators - unseen by the novice user - may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users.
- Allow users to tailor frequent actions.



**Flexibility &
Efficiency of Use**

6.USER CONTROL AND FREEDOM

- Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue.



**User Control
& Freedom**

Class Roster

Academic Year 2023-24 | Semester 2 | University of Glasgow | Postgraduate Taught

COMPSCI 5057 - LC01 (6558)
Human Computer Interaction Design and Evaluation (M) (Lecture)

Days and Times	Room	Instructor	Dates
Fr 12:00-14:00	CTT SIR ALEX STONE:208 LT	Julie Rico	12/01/2024 - 22/03/2024

*Enrollment Status

Enrolled

Enrollment Capacity 141Enrolled 141

Enrolled StudentsPersonalizeFindFirst1-141 of 141Last

Select	ID	Name	Grade Basis	Units	Program and Plan	Level
1	☐	2937025 Adhikari, Mahesh	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
2	☐	2943697 Alkhulaifi, Hala	Non-Graded	0.00	Master of Science(SE) - Software Development, MSc	Postgraduate Level
3	☐	2840472 Bai, Yunfei	Non-Graded	0.00	Master of Science(SE) - Information Technology, MSc	Postgraduate Level
4	☐	2863122 Bi, Sheng	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
5	☐	2938618 CHENG, Xinghan	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
6	☐	2840483 Cai, Yongkang	Non-Graded	0.00	Master of Science(SE) - Software Development, MSc	Postgraduate Level
7	☐	2896869 Chang, Sean	Non-Graded	0.00	Master of Science(SE) - Software Development, MSc	Postgraduate Level

135	☐	2832058 Zhao, Pengfei	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
136	☐	2935107 Zhao, Zexiang	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
137	☐	2841102 Zheng, Cheng	Non-Graded	0.00	Master of Science(SE) - Information Technology, MSc	Postgraduate Level
138	☐	2781134 Zheng, Junnan	Non-Graded	0.00	Master of Science(SE) - MSc in Data Science	Postgraduate Level
139	☐	2877840 Zhu, Jiaqu	Non-Graded	0.00	Master of Science(SE) - Computing Science, MSc	Postgraduate Level
140	☐	2896865 Zhu, Tong	Non-Graded	0.00	Master of Science(SE) - Information Technology, MSc	Postgraduate Level
141	☐	2891886 Zhuo, Yue	Non-Graded	0.00	Master of Science(SE) - Information Technology, MSc	Postgraduate Level

Select All Clear All

Notify Selected Students

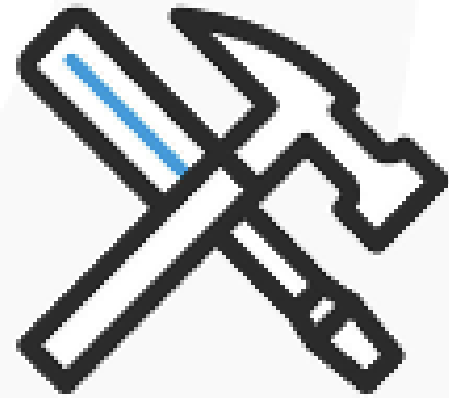
How do you feel when you provide a lot of input to get the results, but find you make one mistake in the input and there is no way to go back to the previous step? v

The importance of the back button!

QUESTION

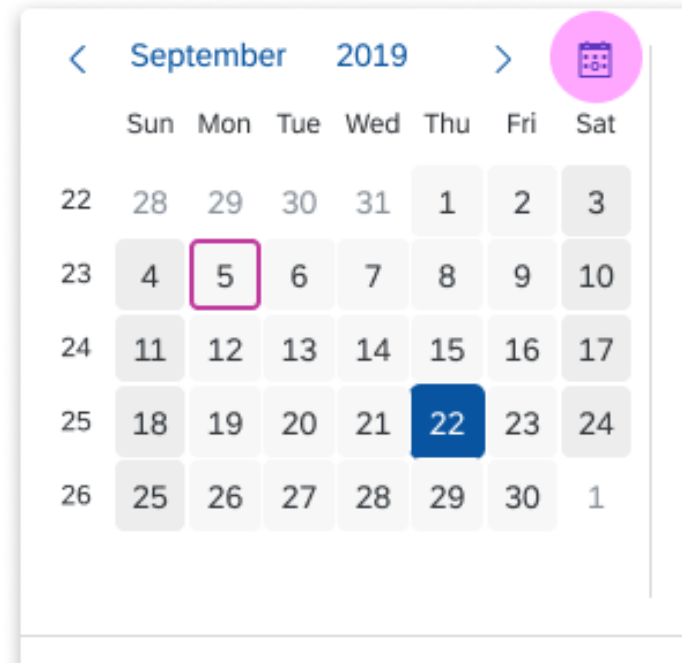
7. ERROR PREVENTION

- Even better than good error messages is a careful design which prevents a problem from occurring in the first place.
- Eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.



Error Prevention

Which is better?



Choose a Start date

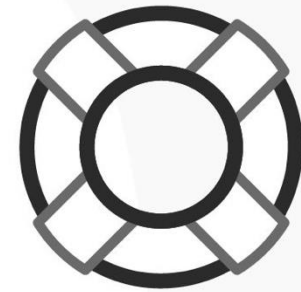
From Date: 10/1/2020 

Please provide a valid Fromdate

QUESTION

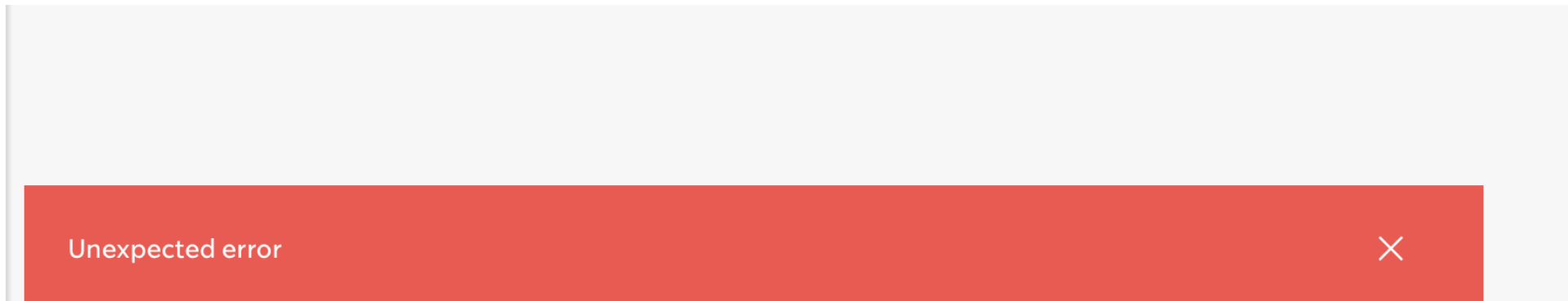
8. HELP USERS RECOGNISE, DIAGNOSE, AND RECOVER FROM ERRORS

- Error messages should be expressed in plain language (no codes)
- Precisely indicate the problem
- Constructively suggest a solution



**Help Users Recognize, Diagnose
& Recover from Errors**

Is the error message helpful?



QUESTION

How can we improve the error message?

Username or email address

thiswasavalidusername

✗ The provided email is invalid

Password

●●●●●●●●

☒ Keep me signed in. [Details](#)

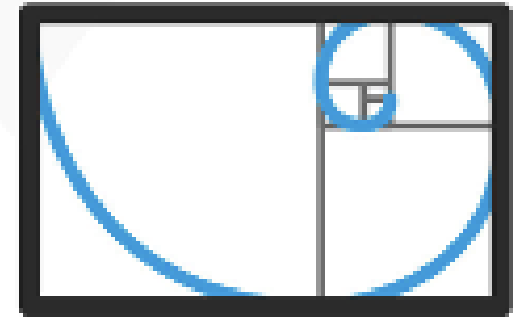
[Forgot your username?](#)

[Forgot your password?](#)

QUESTION

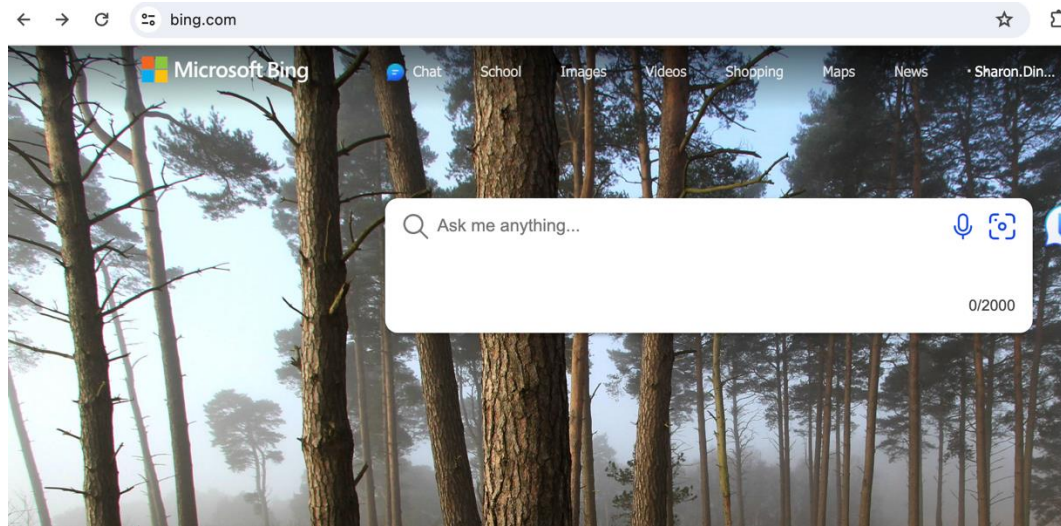
9. AESTHETIC AND MINIMALIST DESIGN

- Dialogues should not contain information which is irrelevant or rarely needed.
- Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.



**Aesthetic &
Minimalist Design**

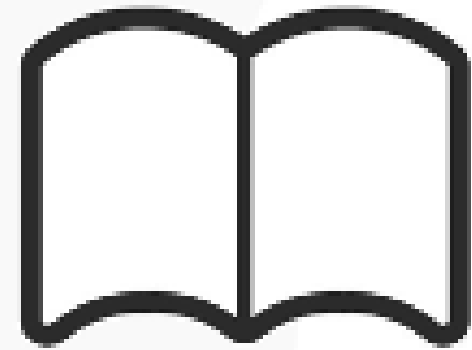
Which you like it better?



QUESTION

10.HELP AND DOCUMENTATION

- It may be necessary to provide help and documentation.
- Information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.



**Help &
Documentation**



HOW TO DO HEURISTIC EVALUATION

Further Reading: <https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>

HEURISTIC EVALUATION

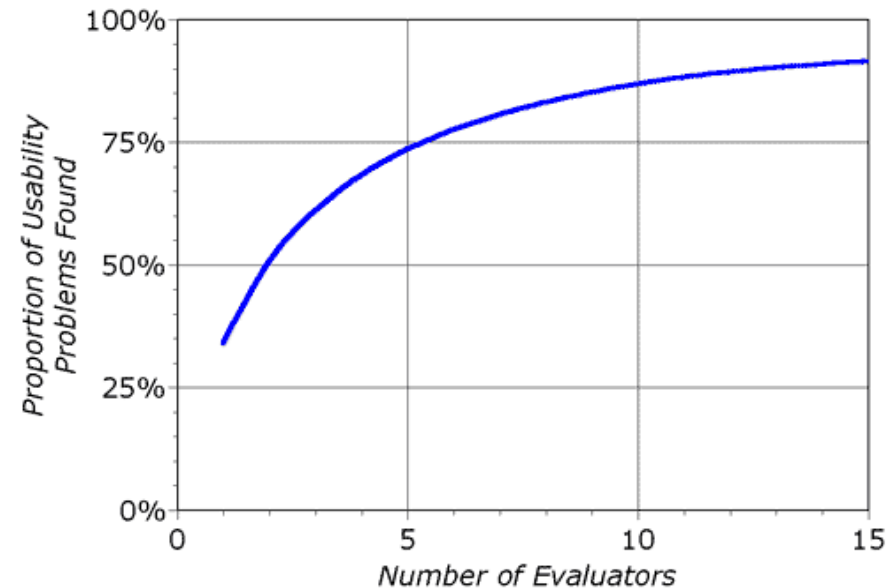
- A usability method uses a small set of evaluators to find usability problems, judged against standard heuristics to provide consistent targets
- Aggregated issues are plotted in a standard format
- Each issue is analysed for the difficulty to correct it and importance

HEURISTIC EVALUATION: EVALUATORS

- Each evaluator finds as many problems as possible alone
- Evaluators find different problems
 - On average, a single evaluator finds only 35% of usability issues identified by a group

HEURISTIC EVALUATION: HOW MANY EVALUATORS

- There are diminishing returns when numerous evaluators are used
- Nielson recommends up to five evaluators, but this should be considered with the cost/benefit of the specific context



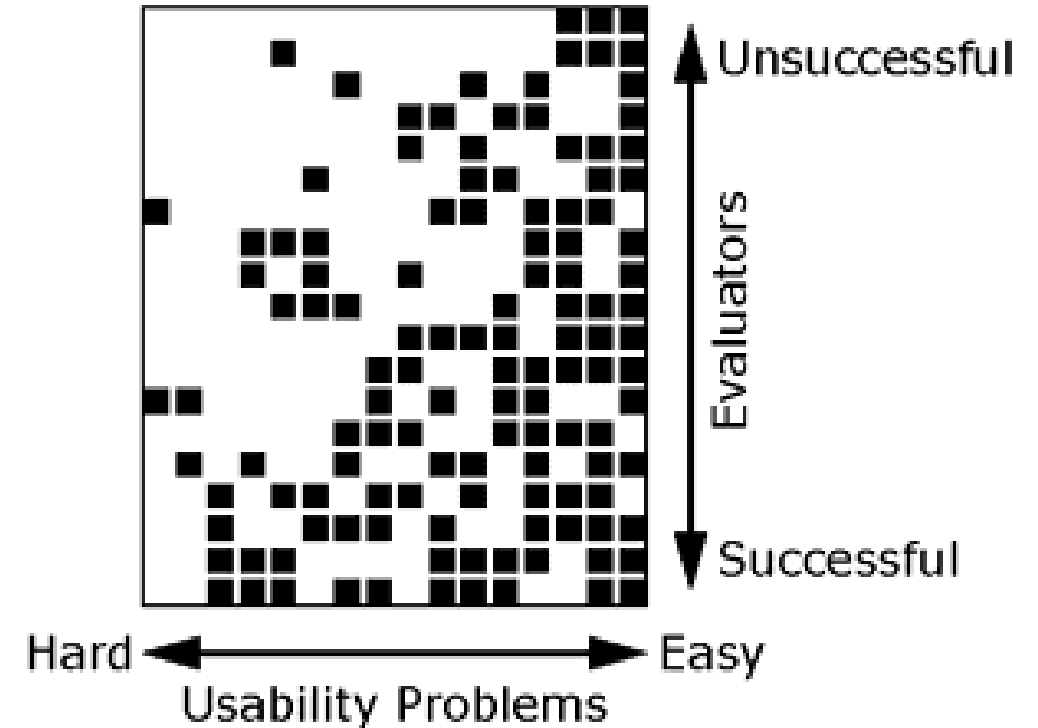
HEURISTIC EVALUATION: IDENTIFYING PROBLEMS

- Problems are presented as a written report
 - Labeled to make it easy for plotting
 - Organized in terms of heuristics

Issue Label	Description	Heuristic	Who
1	XXXX	Visibility of System Status	X, Y, Z
2	XXXX		Z
3	XXXX		Y
4	XXXX	Consistency & Standards	X, Y, Z
5	XXXX	Error Prevention	Y
6	XXXX		Z

HEURISTIC EVALUATION: AGGREGATING PROBLEMS

- A complete list of problems is aggregated from all reports.
- A matrix is created with problems along the x-axis and evaluator along the y-axis
 - This is sorted first by y-axis (successful evaluators) and second by x-axis (difficult to find problems)



HEURISTIC EVALUATION: ANALYSING PROBLEMS

- Consider the prevalence of the problem
 - How many evaluators found it?
 - Consider the ease of correcting the problem
 - Consider the importance of the problem
- Propose solutions and a ranked list of priority